Curriculum Vitae
Dr. Joseph E. Golec

Contact Information

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Research Interests

Experimental Cosmology, Millimeter-Wave Instrumentation, Sunyaev-Zel'dovich Observations, and Galaxy Cluster Astrophysics

Education

University of Chicago, Chicago, IL

Ph.D. Physics — June 2024

Thesis Title: Next-Generation Images of Galaxy Clusters at Millimeter Wavelengths
Advisor: Jeff McMahon

University of Michigan, Ann Arbor, MI

M.S. Physics — 2020

B.S. Physics with Honors & Mathematical Sciences — 2017

Appointments

University of Massachusetts Amherst, Amherst, MA

Postdoctoral Fellow — July 2024 - Present
NSF Astronomy and Astrophysics Postdoctoral Fellow

Department: Astronomy
Scientific Mentor: Grant Wilson
Project: The TolTEC Camera

External Research Funding and Fellowships

NSF Astronomy and Astrophysics Postdoctoral Fellowship — 2024–Present
National Science Foundation
Role: Principal Investigator
Project: AAPF: A Resolved Millimeter-wave Survey of Galaxy Clusters
Award Amount: $330,000

NASA Space Technology Research Fellowship 2019–2023
National Aeronautics and Space Administration
Role: Fellowship Recipient
Project: Optics and Optical Coupling for Future CMB Missions

Research

TolTEC Camera 2019-Present
- Designed and fabricated anti-reflection coatings for silicon optics
- Developed a maximum-likelihood mapmaking pipeline for high-resolution millimeter-wave observations
- Developed an out-of-focus holography pipeline to characterize and improve LMT optical performance
- Led telescope commissioning observations and instrument characterization
- Galaxy Cluster Science Working Group Lead

Experiment for Cryogenic Large-Aperture Intensity Mapping (EXCLAIM) 2020-2024
- Designed metamaterial anti-reflection coating for silicon optics

Cosmology Large Angular Scale Surveyor (CLASS) 2019
- Fabricated metamaterial anti-reflection coating for 150/220 GHz silicon optics

Simons Observatory 2017-Present
- Mass produced metamaterial anti-reflection coating of silicon optics
- Developed and fabricated metamaterial anti-reflection coating for alumina optics

Atacama Cosmology Telescope (ACT) 2013-2022
- Fabricated metamaterial anti-reflection coatings for 20/40 GHz silicon optics
- Coordinated remote telescope operations

Presentations

AAPF Symposium January 2026
An Update on the Commissioning of the TolTEC Camera (Presentation)
Phoenix, AZ

LEPP Journal Club, Cornell University October 2025
TolTEC on the Large Millimeter Telescope: Commissioning Progress and Early Images (Invited Seminar)
Ithaca, NY

Observing the Universe at millimeter wavelengths June 2025
An Update on the Commissioning of the TolTEC Camera (Presentation)
Chicago, IL

Observing the Universe at millimeter wavelengths June 2023

Early High-resolution Millimeter-wave Maps and Instrument Properties of TolTEC (Presentation)
Grenoble, FR

SPIE Astronomical Telescopes + Instrumentation — June 2022
Simons Observatory: Metamaterial Anti-Reflection Coatings for Silicon and Alumina Optics (Poster)
Montreal, QC

New Perspectives — August 2021
Metamaterial Anti-Reflection Coating Alumina Optics for CMB-S4 (Presentation)
Fermilab (Virtual)

SPIE Astronomical Telescopes + Instrumentation — December 2020
Design and fabrication of metamaterial anti-reflection coatings for the Simons Observatory (Poster)
Virtual

Physics Graduate Student Symposium — July 2019
Constraining Neutrino Properties with the Cosmic Microwave Background (Presentation)
University of Michigan

Societies, Honors and Awards

Member of the American Astronomical Society — 2024-Present
Member of the American Physical Society — 2017-Present
Sigma Pi Sigma, Physics Honors Society — 2016
Eagle Scout, Boy Scouts of America — 2013

Teaching

21 cm Mini-School at the University of Massachusetts — 2026
- Led one-day mini-school teaching the fundamentals of radio astronomy with a focus on 21 cm emission
- Developed interactive Python notebooks to teach basics of Galactic 21 cm emission
- Led students in the measurement of Galactic 21 cm emission using a low-cost horn antenna

TA Physics 122 - General Physics II — Winter 2024
University of Chicago
- Taught introductory electricity and magnetism labs to biology and pre-medicine track students
- Led weekly discussion sections to supplement material covered in lecture

Undergraduate Assistant for Michigan Math and Science Scholars — 2017
- Taught special and general relativity to high school students over two weeks as part of a summer camp

- Assisted in the development of the course notes which led to a book publication

Undergraduate Mentoring

Tanvi Bhandurge 2026–Present
Undergraduate Mentee, University of Massachusetts Amherst
Project: Development of instructional materials for an advanced laboratory experiment measuring Galactic 21 cm emission

Amber Duckett 2024–Present
Undergraduate Research Mentee, University of Massachusetts Amherst
Project: Machine learning for out-of-focus holography at the LMT using TolTEC

Professional Service

Served on an NSF peer review panel 2026

Subject-matter expert reviewer in a NASA peer review 2025

Five Colleges Astronomy Department Colloquium Organizing Committee 2024-2025

President of the Society of Physics Students 2016 - 2017
C. Wilber Peters Chapter, University of Michigan

Public Outreach

Astronomy Merit Badge Counselor 2025-Present
- Taught the Astronomy Merit Badge to youth through the Scouting America program

Astronomy is for Everyone 2024 - Present
- Led an initiative on the University of Massachusetts Amherst campus to hold public observing on campus
- Led hundreds of students and the general public in observing with a telescope

Astronomy on Tap 2024 - Present
- Co-hosted events and gave several public lectures
- Ran public observing at Astronomy on Tap events

South Side Science Festival 2023
- Taught the physics of skateboarding to children ages 4-12 from the South Side of Chicago

Publications

First-Author Publications

1. **Golec, J.E.** and the TolTEC Collaboration **March 2024** "Early High-resolution Millimeter-wave Maps From TolTEC" In: EPJ Web Conf., 293 (2024) 00022; doi: 10.1051/epjconf/202429300022

2. **Golec, J.E.**, Sutariya, S., Jackson, R., Zimmerman, J., and Dicker, S.R., et al. **October 2022** "Simons Observatory: broadband metamaterial antireflection cuttings for large-aperture alumina optics" In: Applied Optics, vol. 61, issue 30, p. 8904; doi:10.1364/AO.472459

3. **Golec, J.E.**, McMahon, J.J., Ali, A., Dicker, S., and Galitzki, N., et al. **December 2020** "Design and fabrication of metamaterial anti-reflection coatings for the Simons Observatory" In: Proceedings of the SPIE, Volume 11445, id. 114515T 8 pp.; doi:10.1117/12.2561720

Key Contributor Publications

1. Groth, F., Li, Y., **Golec, J.**, et al. **July 2026** "Constraining Effective Viscosity in the Intracluster Medium via the Thermal Sunyaev-Zeldovich Effect – Predictions from the SLOW Constrained Coma Cluster Simulations" In: eprint arXiv:2607.19753; doi: 10.48550/arXiv.2607.19753

2. Mayerson, D., Charles, A., and **Golec, J.**, **2019** "Relativity: A Journey through Warped Space and Time" In: Springer Briefs in Physics

3. Simon, S.M., **Golec, J.E.**, Ali, A., Austermann, J.E., and Beall, J. et al. **July 2018** "Feedhorn development and scalability for Simons Observatory and beyond" In: Proceedings of the SPIE, Volume 10708, id. 107084B 12 pp.; doi:10.1117/12.2313405

Contributor Publications

1. Essinger-Hileman, T., Ryan, C., Li, Y., et al. **September 2026** "Improved Constraints on Cosmic Microwave Background Circular Polarization with CLASS" In: eprint arXiv:2609.40125; doi: 10.48550/arXiv.2609.40125

2. Dominguez-Frenandez, P., ZuHone, J.A., Silich, E.M., et al. **September 2026** "ICM-SHOX. III. The Case of MACS J0018.5+1626, a Radio Relic That Looks Like a Radio Halo?" In: The Astrophysical Journal, Volume 1008, Issue 1, id.43, 31 pp.; doi: 10.3847/1538-4357/ae8f3c

3. Alford, T., Datta, R., Calabrese, E., et al. **August 2026** "The Atacama Cosmology Telescope: Passband Measurements with an Analysis of Systematic Errors" In: eprint arXiv:2608.18348; doi: 10.48550/arXiv.2608.18348

4. Ma, Z., Wilson, G., Aretxaga, I., et al. **August 2026** "Software framework for TolTEC operation and data automation" In: Proceedings of the SPIE, Volume 14155, id. 141553Y 8 pp. (2026); doi: 10.1117/12.3101313

5. Abril-Cabezas, I., Adachi, S., Ade, P.A.R., et al. **April 2026** "The Simons Observatory: forecasted constraints on primordial gravitational waves with the expanded array of Small Aperture Telescopes" In: Journal of Cosmology and Astroparticle Physics, Volume 2026, Issue 04, id.051, 22 pp.; doi: 10.1088/1475-7516/2026/04/051

6. Aguena, M., Aiola, S., Allam, S., et al. **January 2026** "The Atacama Cosmology Telescope: DR6 Sunyaev-Zel'dovich Selected Galaxy Clusters Catalog" In: The Open Journal of Astrophysics, Volume 9; doi: 10.33232/001c.155863

7. Calabrese, E., Hill, J.C., Jense, H.T., et al. **November 2025** "The Atacama Cosmology Telescope: DR6 constraints on extended cosmological models" In: Journal of Cosmology and Astroparticle Physics, Volume 2025, Issue 11, id.063, 132 pp.; doi: 10.1088/1475-7516/2025/11/063

8. Thibaut, L., La Posta, A., Atkins, Z., et al. **November 2025** "The Atacama Cosmology Telescope: DR6 power spectra, likelihoods and ΛCDM parameters" In: Journal of Cosmology and Astroparticle Physics, Volume 2025, Issue 11, id.062, 101 pp.; doi: 10.1088/1475-7516/2025/11/062

9. Naess, S., Guan, Y., Duivenvoorden, A.J., et al. **November 2025** "The Atacama Cosmology Telescope: DR6 maps" In: Journal of Cosmology and Astroparticle Physics, Volume 2025, Issue 11, id.061, 86 pp.; doi: 10.1088/1475-7516/2025/11/061

10. Venterea, R.C., Orlowski-Scherer, J., Battaglia, N., et al. **August 2025** "The Atacama Cosmology Telescope: Release of A databaSe of millimeTeR ObservatioNs of Asteroids Using acT (ASTRONAUT)" In: eprint arXiv:2508.18300; doi: 10.48550/arXiv.2508.18300

11. Abitbol, M., Abril-Cabezas, I., Adachi, S., et al. **August 2025** "The Simons Observatory: science goals and forecasts for the enhanced Large Aperture Telescope" In: Journal of Cosmology and Astroparticle Physics, Volume 2025, Issue 08, id.034, 45 pp.; doi: 10.1088/1475-7516/2025/08/034

12. Essinger-Hileman, T., Chmaytelli, D., Oxholm, T., et al. **January 2025** "Design and implementation of optics for the experiment for cryogenic large-aperture intensity mapping (EXCLAIM)" In: Review of Scientific Instruments, Volume 96, Issue 1, id.014503, 13 pp.; doi: 10.1063/5.0246814

13. Sierra, C.E., Harrington, K., Sutariya, S., et al. **January 2025** "Simons Observatory: Predeployment Performance of a Large Aperture Telescope Optics Tube in the 90 and 150 GHz Spectral Bands" In: The Astrophysical Journal Supplement Series, Volume 276, Issue 1, id.31, 24 pp.; doi: 10.3847/1538-4365/ad8cdc

14. Dahal, S., Ade, P.A.R., Anderson, C.J., et al. **August 2024** "Superfluid-tight cryogenic receiver with continuous sub-kelvin cooling for EXCLAIM" In: Proceedings of the SPIE, Volume 13102, id. 131022I 13 pp. (2024); doi: 10.1117/12.3018577

15. Coulton, W.R., Madhavacheril, M.S., Duivenvoorden, A.J., Hill, J.C., Abril-Cabezas, I., and Ade, P.A.R., et al. **July 2023** "The Atacama Cosmology Telescope: High-resolution component-separated maps across one-third of the sky" In: eprint arXiv:2307.01258; doi:10.48550/arXiv.2307.01258

16. Madhavacheril, M.S., Qu, F.J., Sherwin, B.D., MacCrann, N., Li, Y., and Abril-Cabezas, I., et al. **April 2023** "The Atacama Cosmology Telescope: DR6 Gravitational Lensing Map and Cosmological Parameters" In: eprint arXiv:2304.05203; doi:10.48550/arXiv.2304.05203

17. Qu, F.J., Sherwin, B.D., Madhavacheril, M.S., Han, D., and Crowley, K.T., et al. **April 2023** "The Atacama Cosmology Telescope: A Measurement of the DR6 CMB Lensing Power Spectrum and its Implications for Structure Growth" In: eprint arXiv:2304.05202; doi:10.48550/arXiv.2304.05202

18. MacCrann, N., Sherwin, B.D., Qu, F.J., Namikawa, T., and Madhavacheril, M.S., et al. **April 2023** "The Atacama Cosmology Telescope: Mitigating the impact of extragalactic foregrounds for the DR6 CMB lensing analysis" In: eprint arXiv:2304.05196; doi:10.48550/arXiv.2304.05196

19. CCAT-Prime Collaboration: Aravena, M., Austermann, J.E., Basu, K., Battaglia, N., and Beringue, B., et al. **January 2023** "CCAT-prime Collaboration: Science Goals and Forecasts with Prime-Cam on the Fred Young Submillimeter Telescope" In: ApJS, Volume 264, Issue 1, id.7, 39 pp.; doi:10.3847/1538-4365/ac9838

20. Chesmore, G.E., Harrington, K., Sierra, C.E., Gallardo, P.A., and Sutariya, S., et al. **December 2022** "Simons Observatory: characterizing the Large Aperture Telescope Receiver with radio holography" In: Applied Optics, vol. 61, issue 34, p. 10309; doi:10.1364/AO.470138

21. Lunde, E., Berthoud, M., DeNigris, N.S., Doyle, S., and Ferrusca, D., et al. **August 2022** "The TolTEC camera: optical alignment and characterization" In: Proceedings of the SPIE, Volume 12190, id. 1219016 6 pp.; doi:10.1117/12.2630340

22. Essinger-Hileman, T., Ade, P., Anderson, C., Barlis, A., and Barrentine, E.M. et al. **August 2022** "EXCLAIM: the EXperiment for Cryogenic Large-Aperture Intensity Mapping" In: Proceedings of the SPIE, Volume 12190, id. 1219009 9 pp.; doi:10.1117/12.2630054

23. McCrackan, M., Ma, Z., DeNigris, N.S., Ryan, C., and Souccar, K., et al. **August 2022** "The TolTEC camera: the citlali data reduction pipeline engine" In: Proceedings of the SPIE, Volume 12189, id. 121891H 16 pp.; doi:10.1117/12.2629095

24. Lungu, M., Storer, E.R., Hasselfield, M., Duivenvoorden, A.J., and Calabrese, E., et al. **May 2022** "The Atacama Cosmology Telescope: measurement and analysis of 1D beams for DR4" In: JCAP Volume 2022, Issue 05, id.044, 55 pp.; doi:10.1088/1475-7516/2022/05/044

25. Dahal, S., Appel, J.W., Datta, R., Brewer, M.K., and Ali, A., et al. **February 2022** "Four-year Cosmology Large Angular Scale Surveyor (CLASS) Observations: On-sky Receiver Performance at 40, 90, 150, and 220 GHz Frequency Bands" In: ApJ Volume 926, Issue 1, id.33, 9 pp.; doi:10.3847/1538-4357/ac397c

26. Switzer, E.R., Barrentine, E.M., Cataldo, G., Essinger-Hileman, T., and Ade, P.A.R., et al. **October 2021** "Experiment for cryogenic large-aperture intensity mapping: instrument design" In: JATIS Volume 7, id. 044004; doi:10.1117/1.JATIS.7.4.044004

27. Zhu, N., Bhandarkar, T., Coppi, G., Kofman, A.M., and Orlowski-Scherer, J.L., et al. **September 2021** "The Simons Observatory Large Aperture Telescope Receiver" In: ApJS Volume 256, Issue 1, id.23, 20 pp.; doi:10.3847/1538-4365/ac0db7

28. Li, Y., Austermann, J.E., Beall, J.A., Bruno, S., and Choi, S.K., et al. **August 2021** "In Situ Performance of the Low Frequency Array for Advanced ACTPol" In: IEEE TASC vol. 31, issue 5, id. 3063334; doi:10.1109/TASC.2021.3063334

29. Mallaby-Kay, M., Atkins, Z., Aiola, S., Amodeo, S., and Austermann, J.E., et al. **July 2021** "The Atacama Cosmology Telescope: Summary of DR4 and DR5 Data Products and Data Access" In: ApJS Volume 255, Issue 1, id.11, 16 pp.; doi:10.3847/1538-4365/abfcc4

30. Xu, Z., Adachi, S., Ade, P., Beall, J.A., and Bhandarkar, T., et al. **April 2021** "The Simons Observatory: The Large Aperture Telescope (LAT)" In: AAS, 5, 100; doi:10.3847/2515-5172/abf9ab

31. Hilton, M., Sifon, C., Naess, S., Madhavacheril, M., and Oguri, M., et al. **March 2021** "The Atacama Cosmology Telescope: A Catalog of >4000 Sunyaev-Zel'dovich Galaxy Clusters" In: ApJS, Volume 253, Issue 1, article id. 3, 25 pp; doi:10.3847/1538-4365/abd023

32. Xu, Z., Chesmore, G.E., Adachi, S., Ali, A., and Bazarko, A., et al. **February 2021** "The Simons Observatory: metamaterial microwave absorber and its cryogenic applications" In: Appl. Opt. 60, 864-874; doi:10.1364/AO.411711

33. Gudmundsson, J.E., Gallardo, P.A., Puddu, R., Dicker, S.R., and Adler, A.E., et al. **February 2021** "The Simons Observatory: modeling optical systematics in the Large Aperture Telescope" In: Appl. Opt. 60, 823-837; doi:10.1364/AO.411533

34. Essinger-Hileman, T., Oxholm, T., Siebert, G., Ade, P., and Anderson, C., et al. **December 2020** "Optical design of the EXperiment for Cryogenic Large-Aperture Intensity Mapping (EXCLAIM)" In: Proceedings of the SPIE, Volume 11453, id. 114530H 15 pp.; doi:10.1117/12.2576254

35. Wilson, G.W., Abi-Saad, S., Ade, P., Aretxaga, I., and Austermann, J., et al. **December 2020** "The TolTEC camera: an overview of the instrument and in-lab testing results" In: Proceedings of the SPIE, Volume 11453, id. 1145302 20 pp.; doi:10.1117/12.2562331

36. Kiuchi, K., Adachi, S., Ali, A., Arnold, K., and Ashton, P., et al. **December 2020** "Simons Observatory Small Aperture Telescope overview" In: Proceedings of the SPIE, Volume 11445, id. 114457L 8 pp.; doi:10.1117/12.2562016

37. Cataldo, G., Ade, P.A.R, Anderson, C.J., Barlis, A., and Barrentine, E.M., et al. **December 2020** "Overview and status of EXCLAIM, the experiment for cryogenic large-aperture intensity mapping" In: Proceedings of the SPIE, Volume 11445, id. 1144524 11 pp.; doi:10.1117/12.2561069

38. Aiola, S., Calabrese, E., Maurin, L., Naess, S., and Schmitt, B.L., et al. **December 2020** "The Atacama Cosmology Telescope: DR4 maps and cosmological parameters" In: JCAP 12 047; doi:10.1088/1475-7516/2020/12/047

39. Choi, S.K., Hasselfield, M., Ho, S.P., Koopman, B., and Lungu, M., et al. **December 2020** "The Atacama Cosmology Telescope: a measurement of the Cosmic Microwave Background power spectra at 98 and 150 GHz" In: JCAP 12 045; doi:10.1088/1475-7516/2020/12/045

40. Yagoubov, P., Mroczkowski, T., Belitsky, V., Cuadrado-Calle, D., and Cuttaia, F., et al. **February 2020** "Wideband 67-116 GHz receiver development for ALMA Band 2" In: A&A 634, A46; doi:10.1051/0004-6361/201936777

41. Ade, P., Aguirre, J., Ahmed, Z., Aiola, S., and Ali, A., et al. **February 2019** "The Simons Observatory: science goals and forecasts" In: JCAP 1902 056; doi:10.1088/1475-7516/2019/02/056

42. Galitzki, N., Ali, A., Arnold, K.S., Ashton, P., and Austermann, J.E., et al. **July 2018** "The Simons Observatory: instrument overview" In: Proceedings of SPIE, Volume 10708, pgs 10708 - 10708 - 13; doi:10.1117/12.2312985